D Y PATIL INTERNATIONAL UNIVERSITY
AKURDI PUNE

Embedded System Design and Development

Project Report.

Title: Car Parking System Using Arduino

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Introduction:

Need for the project

This system solves several common and frustrating problems associated with small, unmanaged parking lots:

1. Time and Fuel Waste: In busy areas, drivers waste significant time and fuel circling a parking lot searching for a space that may not exist.
2. Traffic Congestion: The "hunting" behaviour of drivers looking for spots creates unnecessary traffic congestion within the parking lot, leading to delays and potential accidents.
3. Driver Frustration: The uncertainty of finding a spot, especially at a small clinic, office, or apartment building, creates a poor experience for visitors and tenants.
4. Inefficient Use of Space: Without a system, there is no way for a lot manager (or other drivers) to know if the lot is being used at its full capacity.

This project directly solves the problem of uncertainty by providing clear, immediate information to drivers before they even enter the lot.

Purpose

The primary goal of this system is to automate and manage access to a small parking lot by keeping a real-time count of available spaces.

To achieve this goal, the system has several key purposes:

1. To Track Occupancy: To use an entry sensor and an exit sensor to accurately increment and decrement a counter (available slots).
2. To Control Access: To use a servo motor as a gate, physically preventing entry when the lot is full (availableSlots == 0).
3. To Inform Drivers: To use an LCD screen to clearly display the lot's status (e.g., "Slots: 2/4" or "Sorry, Lot is Full.").
4. To Improve Flow: To open the gate automatically for entering cars (when space is available) and allow free exit, streamlining the parking process.

Usefulness

This system provides clear benefits to three distinct groups:

For Drivers (Users):

1. Saves Time: Drivers are instantly informed if a lot is full, allowing them to seek alternatives immediately instead of wasting time.
2. Reduces Stress: It eliminates the guesswork and frustration of searching for a parking spot.

For Lot Owners (e.g., small businesses, apartments):

1. Automation: It provides 24/7 management of the lot without needing a human attendant, saving on labour costs.
2. Efficiency: It ensures the lot is used to its maximum capacity without becoming over-full, which can cause chaos.
3. Professionalism: It adds a modern, helpful feature to the property, improving the experience for customers or residents.

For Society & the Environment:

1. Reduces Emissions: By eliminating unnecessary circling, the system helps reduce vehicle emissions and fuel consumption on a local scale.
2. Improves Local Traffic: It prevents cars from queuing up to enter a full lot, which can help reduce congestion on adjacent streets.

Reliability

As a prototype built with the specified components, the system is reliable for demonstrating the concept, but it has known vulnerabilities in a real-world scenario:

1. Sensor Accuracy: IR (Infrared) sensors can be affected by bright, direct sunlight, which can "blind" them. They can also be triggered by things other than cars, like a person walking very close to them.
2. Tailgating: The current code (while(digitalRead(entrySensorPin) == LOW)) waits for the first car to pass. If a second car "tailgates" the first one before the gate closes, it might enter without being counted, leading to an inaccurate availableSlots count.
3. Power Loss: If the Arduino loses power, the availableSlots variable will reset to its initial value (4), even if the lot was full. This would require a manual reset.
4. Component Failure: The servo motor, as a mechanical part, has a limited lifespan and could fail (e.g., get stuck open or closed).

To improve reliability for a commercial product, one would upgrade the IR sensors to more robust ones (like ultrasonic sensors or magnetic loops in the pavement) and add a way to save the count to permanent memory (EEPROM) in case of a power failure.

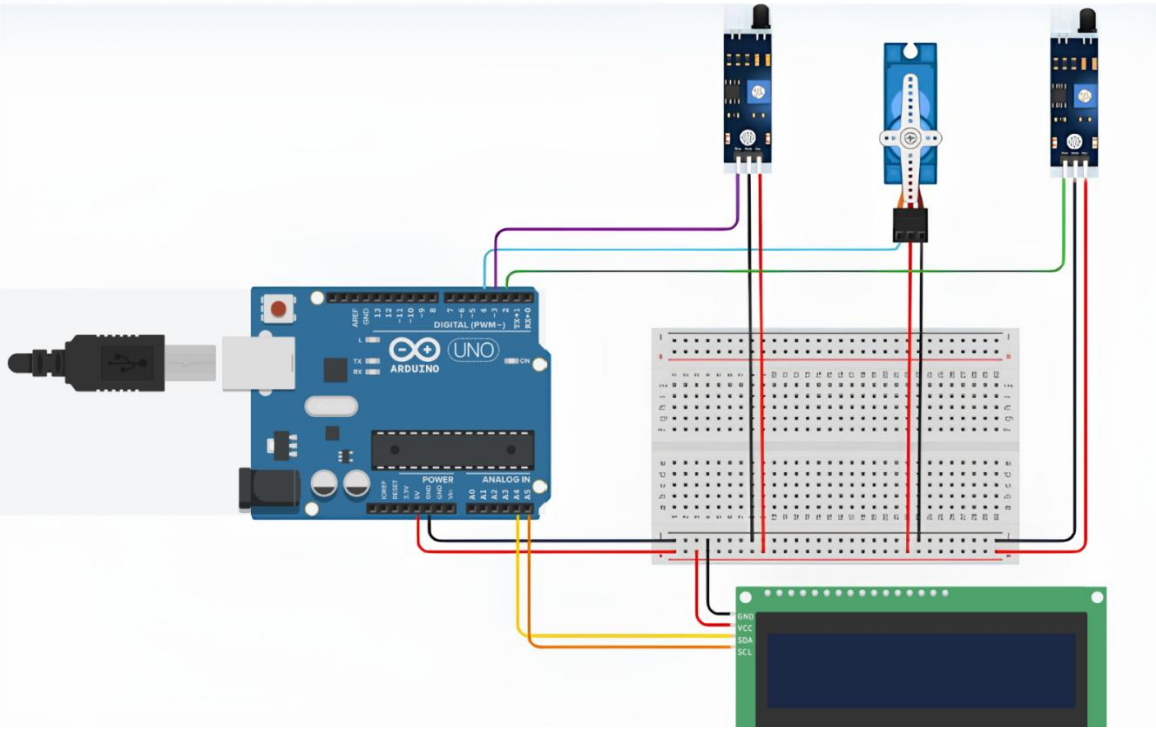
Innovation

The innovation of this project lies in its simplicity, affordability, and scalability-down for a market that is often ignored.

1. No System (Dumb Lot): It is infinitely better than a simple "Parking" sign, as it provides crucial real-time data.
2. Human Attendant: It is far more cost-effective for a small 4-space lot, works 24/7 without breaks, and is not prone to human error.
3. Large-Scale Systems: Massive commercial systems (like at airports) use expensive per-spot sensors, complex networks, and custom software. This project is innovative because
4. achieves a similar gate-level result (telling you if the lot is full, not which spot) for a tiny fraction of the cost.

Its key innovation is making "smart" technology accessible and practical for very small-scale applications, which are currently unserved by expensive, large-scale commercial solutions.

Circuit Diagram:




```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
LiquidCrystal_I2C lcd(0x27,16,2);
#include <Servo.h>
Servo myservo;
int IR1 = 2;
int IR2 = 3;
int Slot = 4; //Total number of parking Slots
int flag1 = 0;
int flag2 = 0;
void setup() {
  Serial.begin(9600);
    lcd.init(); //initialize the lcd
    lcd.backlight(); //open the backlight
pinMode(IR1, INPUT);
pinMode(IR2, INPUT);
myservo.attach(4);
myservo.write(100);
lcd.setCursor (0,0);
lcd.print("     ARDUINO     ");
lcd.setCursor (0,1);
lcd.print(" PARKING SYSTEM ");
delay (2000);
lcd.clear();
}
void loop(){
if(digitalRead (IR1) == LOW && flag1==0){
if(flag2==0){myservo.write(0); Slot = Slot-1;}
}else{
lcd.setCursor (0,0);
lcd.print("    SORRY :(    ");
lcd.setCursor (0,1);
lcd.print("  Parking Full  ");
delay (3000);
lcd.clear();
}
}
if(digitalRead (IR2) == LOW && flag2==0){flag2=1;
if(flag1==0){myservo.write(0); Slot = Slot+1;}
}
if(flag1==1 && flag2==1){
delay (1000);
myservo.write(100);
flag1=0, flag2=0;
}
lcd.setCursor (0,0);
lcd.print("    WELCOME!    ");
lcd.setCursor (0,1);
lcd.print("Slot Left: ");
lcd.print(Slot);
}
```

List of Hardware Components: For Eg:

sr.	Component name	Specification / model	quantity	Purpose / function
1	Arduino Uno	microcontroller board	1	This is the "brain" that runs your code to read sensors, control the servo, and send text to the LCD.
2	IR Sensor	A small module with an infrared light transmitter and receiver pair.	2	To detect an object by sending out an IR beam and sensing if it reflects back; in your project, it's used to tell if a car is present at the gate or exit.
3	SG90 Micro Servo motor	A lightweight, 9-gram (9G) micro servo motor that rotates 180 degrees.	1	This is the **physical gate arm** in your project; it's a motor that moves to a precise angle (like 0° to 90°) when commanded by the Arduino.
4	16x2 LCD Display with I2C Module	A 16-character by 2-line text display with an I2C serial interface "backpack."	1	To show information (like "Slots: 4/4" or "Lot is Full") using only two data pins from the Arduino.

Working Principle:

1. Arduino Uno (Microcontroller)

- Acts as the brain of the system.
- Continuously reads signals from sensors (entry and exit).
- Executes logic to increment/decrement the slot counter.
- Controls the servo motor (gate) and updates the LCD display.
- Provides real-time decision-making based on available slots.

2. IR Sensors (Entry & Exit Detection)

- Entry Sensor: Detects when a car approaches the gate. Sends a HIGH/LOW signal to Arduino.
- Exit Sensor: Detects when a car leaves the lot.
- Working principle: IR transmitter emits infrared light, and the receiver detects reflection changes when a car interrupts the beam.

3. Servo Motor (Gate Control)

- Functions as the automatic gate barrier.
- Controlled by PWM (Pulse Width Modulation) signals from Arduino.
- Rotates to open when a car is allowed entry, and returns to closed position afterward.
- Prevents cars from entering when the lot is full.

4. LCD Display (Driver Information)

- Displays the current parking status (e.g., “Slots: 2/4” or “Lot Full”).
- Works on the principle of converting electrical signals from Arduino into visible characters.
- Provides instant feedback to drivers before they enter.

5. Power Supply

- Provides stable 5V DC power to Arduino, sensors, and servo motor.
- Ensures uninterrupted functioning of the system.
- Can be from USB, adapter, or battery pack.

6. Resistors & Supporting Circuitry

- Used with IR sensors to limit current and ensure proper signal levels.
- Maintain stability and protect components from damage.

7. Connecting Wires & Breadboard

- Provide physical connections between components.
- Breadboard allows easy prototyping without soldering.
- Ensures signals and power are routed correctly.

Together, these components form a closed-loop system:

- Sensors detect → Arduino processes → Servo motor acts → LCD informs → Power keeps it running.

Applications of the Module:

1. Residential: Apartment complexes, condominiums, and gated communities to manage private parking for residents and visitors.
2. Commercial & Corporate: Small office buildings or corporate campuses to manage employee or guest parking lots.
3. Retail & Hospitality: Restaurants, cafes, small hotels, and motels with limited, dedicated customer parking. Healthcare: Small clinics, dental offices, or veterinary practices to ensure parking is available for patients.
4. Public Services: Small government offices, libraries, or community centres with their own parking lots.
5. Event Management: To control access to small, temporary, or VIP parking areas during events.